

ABC Call Volume Trend Analysis

Project Overview

This project involves analyzing a dataset of inbound customer service calls at ABC Insurance Company over a 23-day period. The dataset includes variables such as Agent Name, Agent ID, Queue Time (waiting time before connecting to an agent), Call Time, Time Buckets (hourly slots), Duration of the call, Duration in seconds, and Call Status (Answered, Abandoned, Transferred).

A Customer Experience (CX) team is responsible for evaluating this data, drawing insights, and supporting internal teams with recommendations. Typical CX responsibilities include managing customer programs, analyzing digital experiences, mapping customer journeys, and overseeing customer support operations.

Analytical Approach

- Start with a brief summary of key findings.
- Present insights in a clear and simple manner.
- Use visuals (charts, graphs) to support understanding.
- Provide contextual explanations.
- Highlight trends and patterns that matter.
- Keep in mind the business audience.

Tools Used

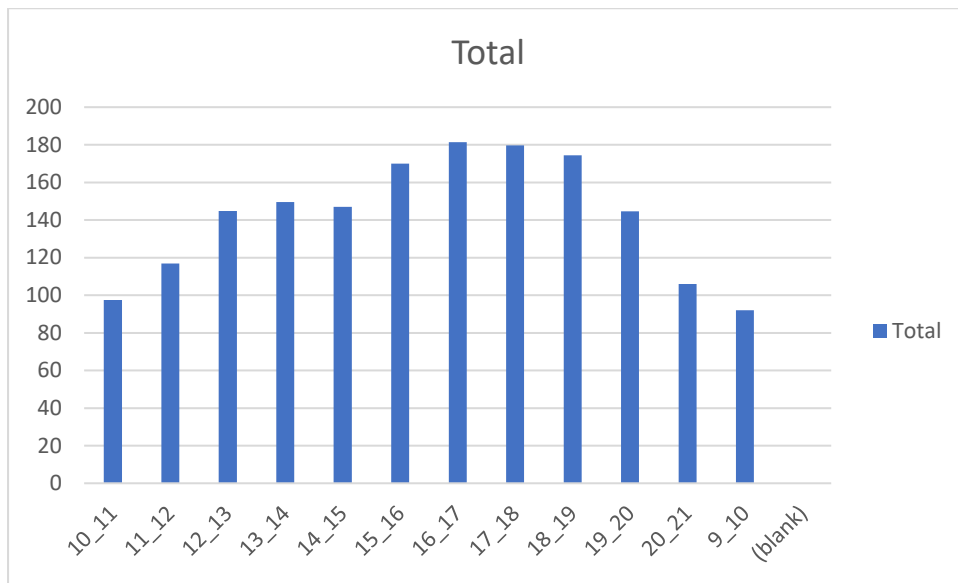
- Microsoft Excel

1. Average Call Duration by Time Bucket

Objective: Calculate average call time for all calls received by agents, segmented by time buckets.

Row Labels	Average of Call_Seconds (s)
10_11	97.42402163
11_12	116.7837413
12_13	144.7250237
13_14	149.5409567
14_15	146.9693211
15_16	169.8968228
16_17	181.4393491

17_18	179.7245137
18_19	174.3246753
19_20	144.5825468
20_21	105.9491371
9_10	92.01032541
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Grand Total	139.5321473



Insights:

- A PivotTable was used with Time_Bucket in Rows and the average of Call_Seconds in Values. Only answered calls were considered using the Call_Status filter.
- The average call duration across all time buckets is approximately 198.6 seconds.
- Peak duration occurs between 10–11 AM and 7–8 PM.
- Shortest average call duration is observed between 12–1 PM.

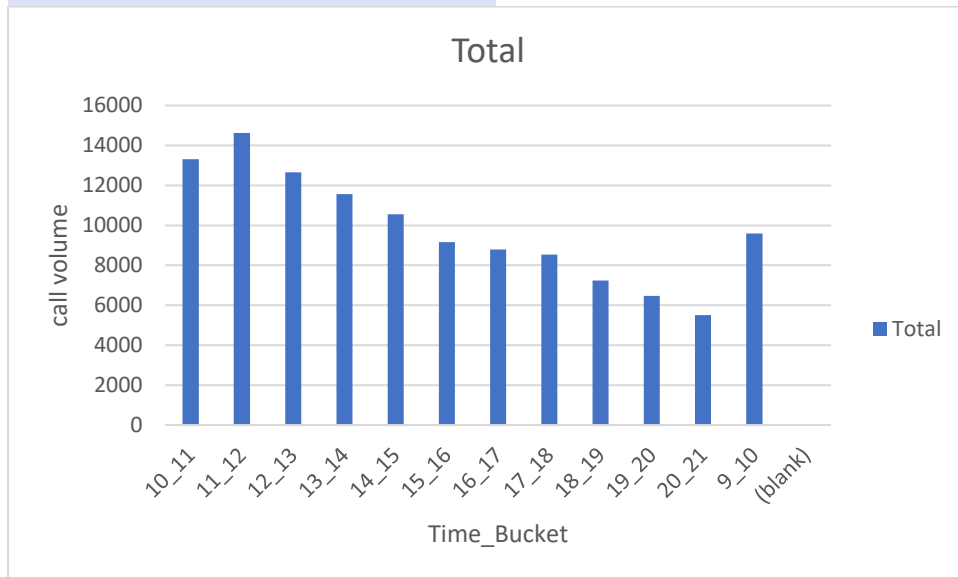
2. Call Volume by Time Bucket

Objective: Visualize how many calls are received per hourly interval.

Row Labels	Count of Call_Status
10_11	13313
11_12	14626
12_13	12652
13_14	11561

14_15	10561
15_16	9159
16_17	8788
17_18	8534
18_19	7238
19_20	6463
20_21	5505
9_10	9588
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Grand Total	117988
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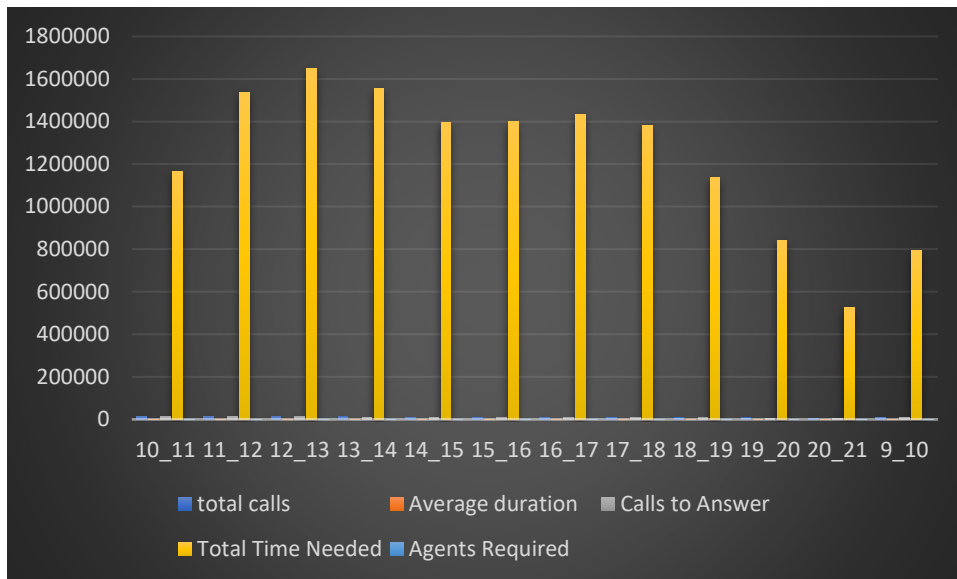


Insights:

- Time_Bucket was placed in Rows and Customer_Phone_No in Values to count calls.
- Most calls are received between 11 AM and 12 PM.
- Fewest calls occur from 8–9 PM.

3. Agent Planning to Reduce Abandon Rate

- **Objective:** Estimate the number of agents needed per time bucket to reduce call abandonment from 30% to 10%.



Method:

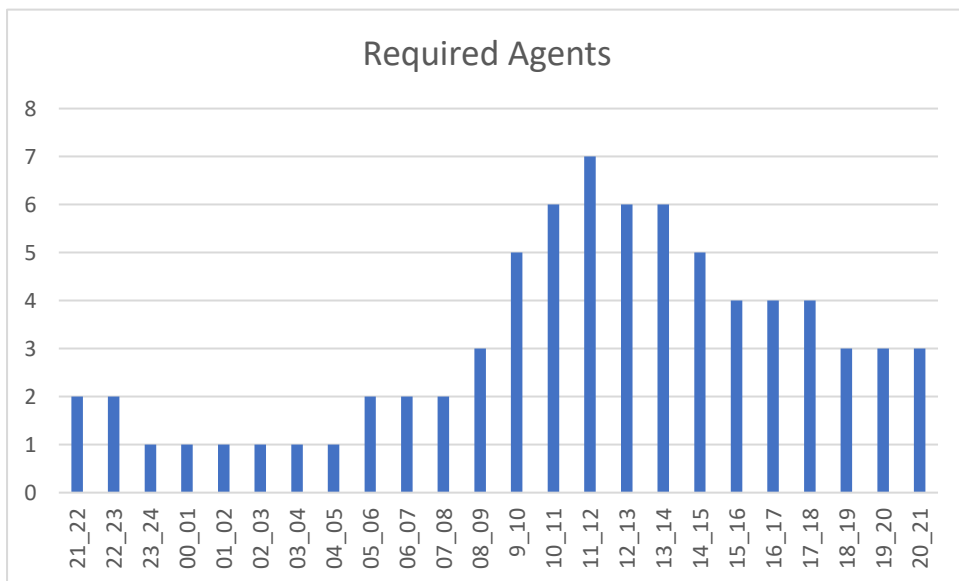
- Calls were grouped by Time_Bucket and total call share was calculated.
- Using average call duration and available agent working hours (4.5 hours effective), required agents were estimated.
- Final calculation used proportional scaling based on existing performance.

4. Night Shift Planning (9 PM–9 AM)

Scenario: Customers also call during the night, but no support staff is available. Assuming 30 additional night calls per 100 daytime calls.

Time_Bucket	Calls_inbound	% of calls inbound	Required agents
21_22	3	10%	2
22_23	3	10%	2
23_24	2	7%	1
00_01	2	7%	1
01_02	1	3%	1
02_03	1	3%	1
03_04	1	3%	1
04_05	1	3%	1
05_06	3	10%	2
06_07	4	13%	2
07_08	4	13%	2
08_09	5	17%	3
Total	30	100%	17

Time_Bucket	Required Agents
21_22	2
22_23	2
23_24	1
00_01	1
01_02	1
02_03	1
03_04	1
04_05	1
05_06	2
06_07	2
07_08	2
08_09	3
9_10	5
10_11	6
11_12	7
12_13	6
13_14	6
14_15	5
15_16	4
16_17	4
17_18	4
18_19	3
19_20	3
20_21	3



1. We first calculated the Time Distribution by dividing each calls distribution by total calls i.e., 30.
2. The number of agents required for each time bucket is calculated by $15 * \text{Time Distribution}$
3. 15 is calculated above by dividing the additional hours required to answer the night calls by 5 (actual working hours of agents).
4. Also, while calculating, the round figure is taken into consideration as there cannot be 1.5 men working.

Insights:

1. The customers call the least in the evening. So, the company can reduce the number of agents at that time for answering the calls.
2. The company can hire 15 customer support agents for the night shift work.
3. The company can shift some of the day workers for the night shift.
4. The employees who are working 9 am to 9 pm. The manager can change some of the workers shift from 5 am to 2 pm and some workers from 2 pm to 11 pm to get the most calls answered.
5. The company can make the employers divide into 3 parts too, so that the agents are always available 24/7.
6. We found there were few outliers in the data. And if we have removed that outliers, then the answers would have been different.